MELBOURNE AIRBNB LAUNCH STRATEGY REPORT

Subject Code: CMCE30005	**Subject Name:** Business Analytics Challenge
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Tutor Day/Time: Thursday, 10am-12pm	**Tutor Name:** Jeanette Wang
Assignment Name: Final Report	
Due Date: 23rd October 2026	
Word Count: 2816 **GitHub repository:** https://github.com/Qihang-cpu/CMCE30005-Renascence **AI Declaration:** We declare that the submission is original and solely our work, all sources and tools used have been appropriately cited or acknowledged, and any copied or modified code has been declared using the methods indicated by our subject coordinator.	

Introduction

We will act as business analytics consultant for our client who intends to list multiple properties on Airbnb in Melbourne. We make no assumptions about the types of properties our clients own, so “properties” includes all types, such as houses and apartments. Accordingly, we have formulated the following research questions: for new operator preparing to list or act as agents for Airbnb listings in Melbourne, how can they select appropriate rental products based on market data and determine the initial listing price and operation plan. To address the business problem, we focus on four key questions: which is the best region, property type, listing price, and operational planning. Each question will be examined in detail in the following sections. Since Airbnb Superhosts must complete multiple bookings and maintain high ratings, it is likely that they are experienced operators who are less likely to seek professional advice. Therefore, we only focus on non-superhosts, whom we assume to be “new operators”. The project examines existing properties or those under consideration for management, without estimating returns on property acquisition.

Data Understanding and Methods

We used data from Inside Airbnb snapshot of Greater Melbourne, which mainly consists of three parts: listings, calendar and reviews. The listings dataset contains property details, including location, type, capacity, and amenities, we use it for market comparison and price prediction. The calendar and reviews datasets provide dates and guest feedback, those are used to calculate stay duration and analyze guest experiences respectively.

To support the presentation of complex charts, the entire analysis was conducted in Python. Analysis is organized into three stages: descriptive, predictive, and prescriptive. Descriptive analysis identifies the most

suitable property types and locations, while predictive analysis determines the optimal listing price. Prescriptive analysis will integrate these findings to an optimal operational recommendation plan. Due to the right skew pattern in listing prices, ridge regression with a log-transformed target is considered first. Random forest and gradient boosting regression are also evaluated to capture potentially nonlinear relationships between listed price and final transaction price. The best-performing model is then selected from these three candidates.

Data Cleaning and EDA Process

We used the listing as the main table and associated the review with the calendar according to the property number. To reflect general affordability, listing prices were restricted to AUD 20–2,000. The analysis included properties with stays longer than at least one week and one month history of reviews to ensure valid data. Moreover, as we posted above, we only consider non-superhost data. Superhosts were retained in the original dataset but excluded from the research sample and subsequent core analysis and price prediction modelling. 25.47% of listed accommodation price for the properties in listings is missing. Due to a relatively high proportion, missing values are retained for supply statistics but excluded from the price model.

After the entire data cleaning process, the final sample comprised 5,673 independent properties for price analysis and prediction, with 8,214 entire-home listings retained for overall description of the market. According to the regional division of the local government of Victoria, we assume the area of “Inner Metro” as the city center, while the areas outside, such as “Eastern Metro”, are assumed as the suburbs (Department of Transport and Planning, n.d.). Variable “area” was derived from the original “neighbourhood” variable to group Melbourne properties sample by a more detailed region scale to support more precise operational plans.

Moreover, the inconsistent property classifications in the original data are standardized by a new variable "property_group". It classifies properties into four categories: apartment, house/townhouse, cottage/guesthouse, and other. This will help us to analyze the best property types based on the best region.

Analysis and Results

Descriptive analysis

This process examines differences in listing prices across property types and regions under current market conditions, providing a foundation for analysis and conclusions. We use the 8,214 entire-home listings that were grouped by "area" to produce a bar chart of listing counts by area (Figure 1) and a map of listing locations (Figure 2) for market region analysis.

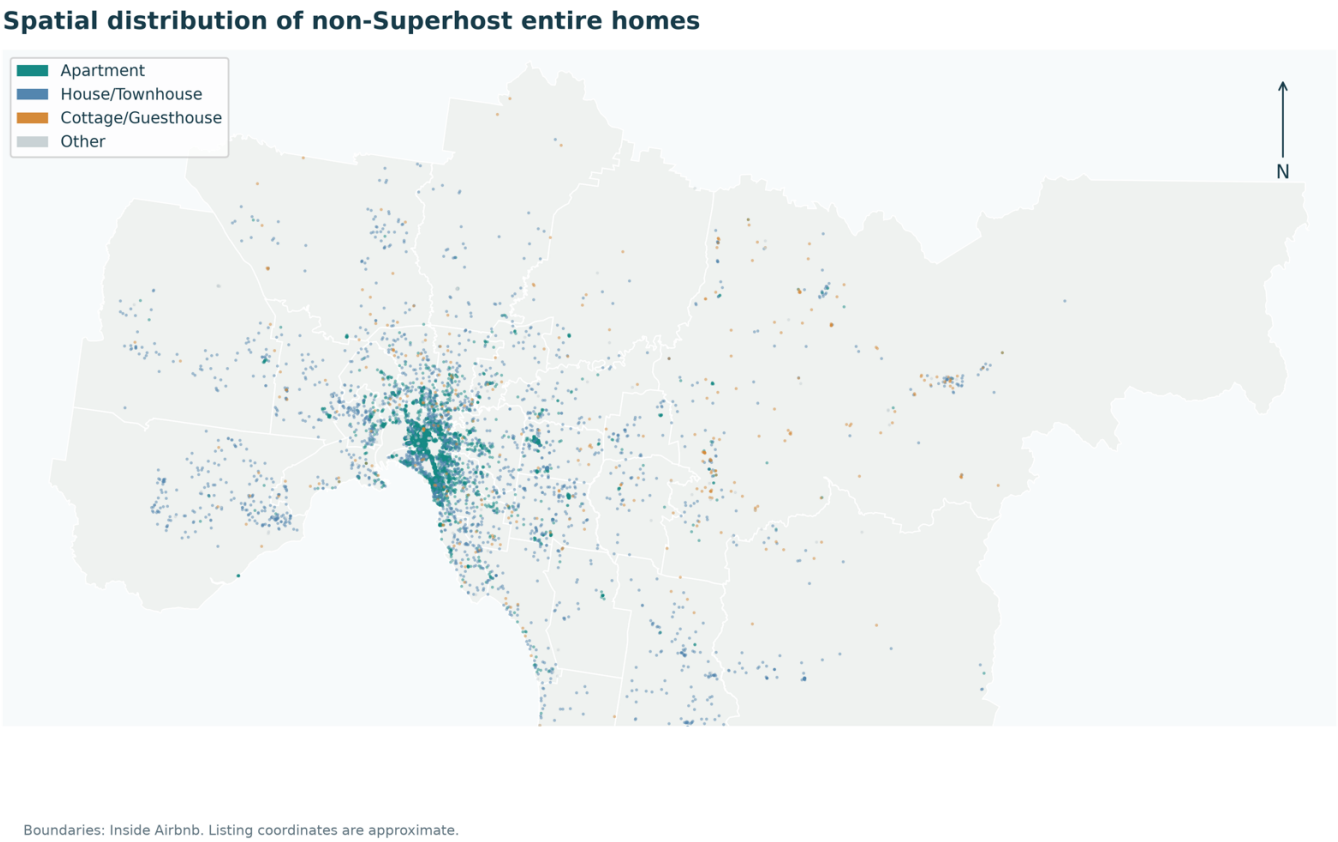


Figure 1

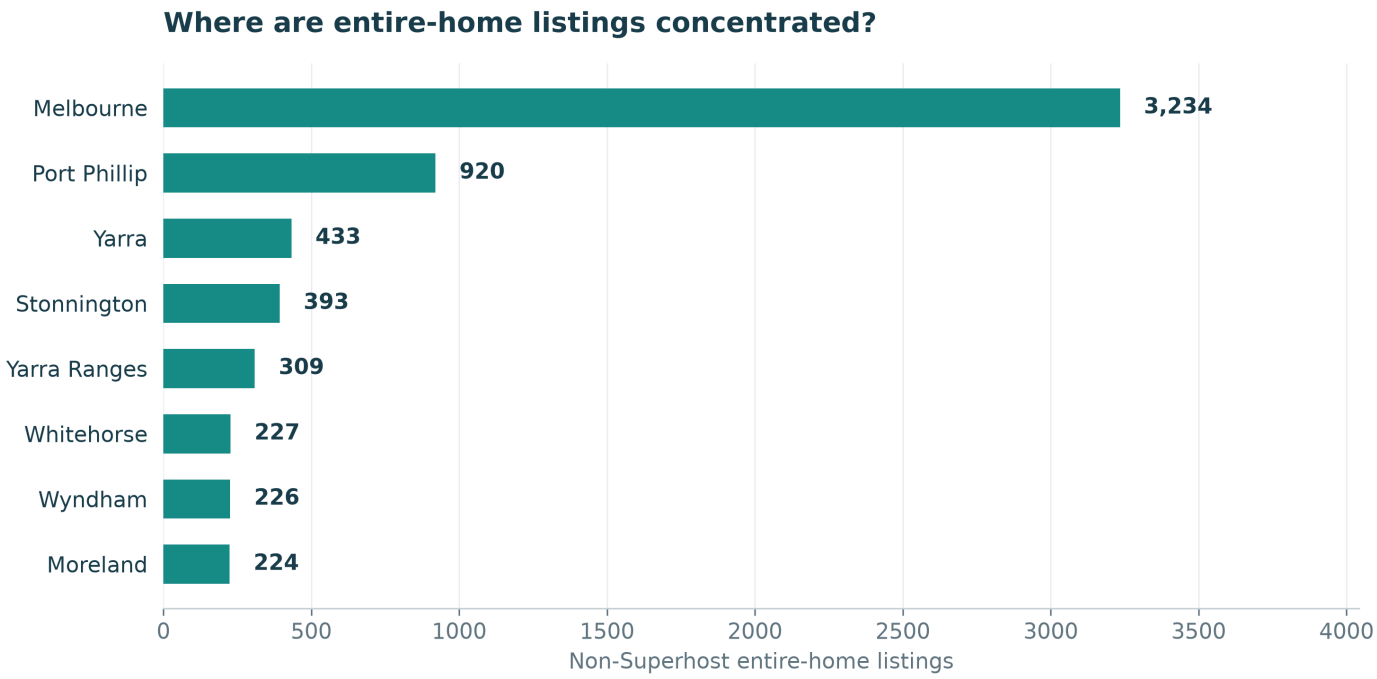


Figure 2

Based on the outcome of those two figures, the properties are predominantly concentrated in central Melbourne, with a more dispersed distribution across suburban areas. Due to the high concentration of listings in central Melbourne, subsequent comparisons should be conducted by region and property types to avoid applying city-centre-dominated findings to all regions. Therefore, in the next step, listings were cross tabulated by “area” and “property_group” and the proportion of each property type within different regions will be calculated to find the best property. The outcome shows in the area near centre of Melbourne, the sample listings are mainly apartments (accounting for more than 71%), while in the suburbs such as Yarra range, houses/townhouses and cottages/guesthouses occupy a dominant position (both accounting for 41%). Hence, there is an obvious regional and product differences. The next analysis should be conducted by distinguishing between the urban apartment market and the peripheral residential and leisure accommodation markets.

The estimated annual income variable "estimated_revenue_l365d" is used as our criterion for evaluating the best properties. Higher revenue could be associated with high scored listings because they may countian better

bedroom count, guest capacity, or location. Therefore, we used a multiple regression model to examine the association between higher review score and higher income, with "review_scores_rating" as the main independent variable, "estimated_revenue_l365d" as the dependent variable, and other property characteristics, such as the number of bedrooms, as control variables. The output shows coefficient of the review to revenue is 12,531.19, with the p-value is less than 0.001. Therefore, we reject the null hypothesis at the 5% significance level. The positive coefficient indicates a statistically significant positive association between ratings and estimated annual revenue, after controlling for other variables in the model. Furthermore, bedrooms are positively associated with estimated annual revenue, while minimum stay is negatively associated. All three associations are statistically significant, holding other variables constant.

The output of the descriptive analysis indicates the most suitable areas should be assessed separately. Long-term rental properties with more bedrooms could serve as a starting point for property selection, while improving guest ratings could be considered an actionable recommendation.

Predictive Analysis

This process will select the best from three models: ridge regression, random forest and gradient boosting regression. The best starting listing price will be determined, and all models will use a unified independent variable and dependent variable. Moreover, this process will use a sample comprised 5,673 independent properties for price analysis and prediction, which is selected based on the filtering conditions we mentioned in data cleaning part.

The model takes the nightly listed price as the dependent variable, and all other variables such as region, room type, number of bedrooms, and available occupancy as predictor variables. Our listing price prediction model

targets properties that are about to enter the market and typically do not yet have guest ratings. So variable review will not be considered in our predictive model. We select the model based on the lowest mean absolute error (MAE) of the validation set, and the outcome shows gradient boosting regression has the lowest MAE (74.89 < 76.10 < 79.36). This might because this model can gradually increase the decision tree and correct the previous prediction errors. So gradient boosting regression is our final model. Building a model does not necessarily mean that it can provide an accurate quotation, we build another test to prove the accuracy of the model. A simple pricing baseline—the median listing price of properties with the same bedroom count in the same area—will be compared with the model price. The method with the lowest test-set MAE will be selected to provide clients with more accurate listing price estimates. The outcome also indicates our model have better prediction of the listing price (MAE = 87.78 < 93.74).

Three priority products are selected using a minimum sample threshold and review activity rankings, then assigned reference prices by the chosen prediction model. Based on the conclusion of the descriptive analysis, we grouped the samples using the grouping variables “area x property type x number of bedrooms”, and finally obtained 262 product groups. Listings were further screened by comparable listing counts, distinct host counts, listings with at least one year of review history, competitors offering valid quotes and permitting long stays, and the availability of model-generated reference prices. Multiple screening rounds applied progressively stricter thresholds to assess reliability. Given the positive association between reviews and income, the three screened property groups with the highest median counts of annual high reviews were selected as the top recommendations. Although the analysis was conducted by region, suburban property groups had fewer high ratings than those in the city center and therefore did not rank among the top three. Therefore, we used the model to predict their listed prices, and the results are shown in Figure 3.

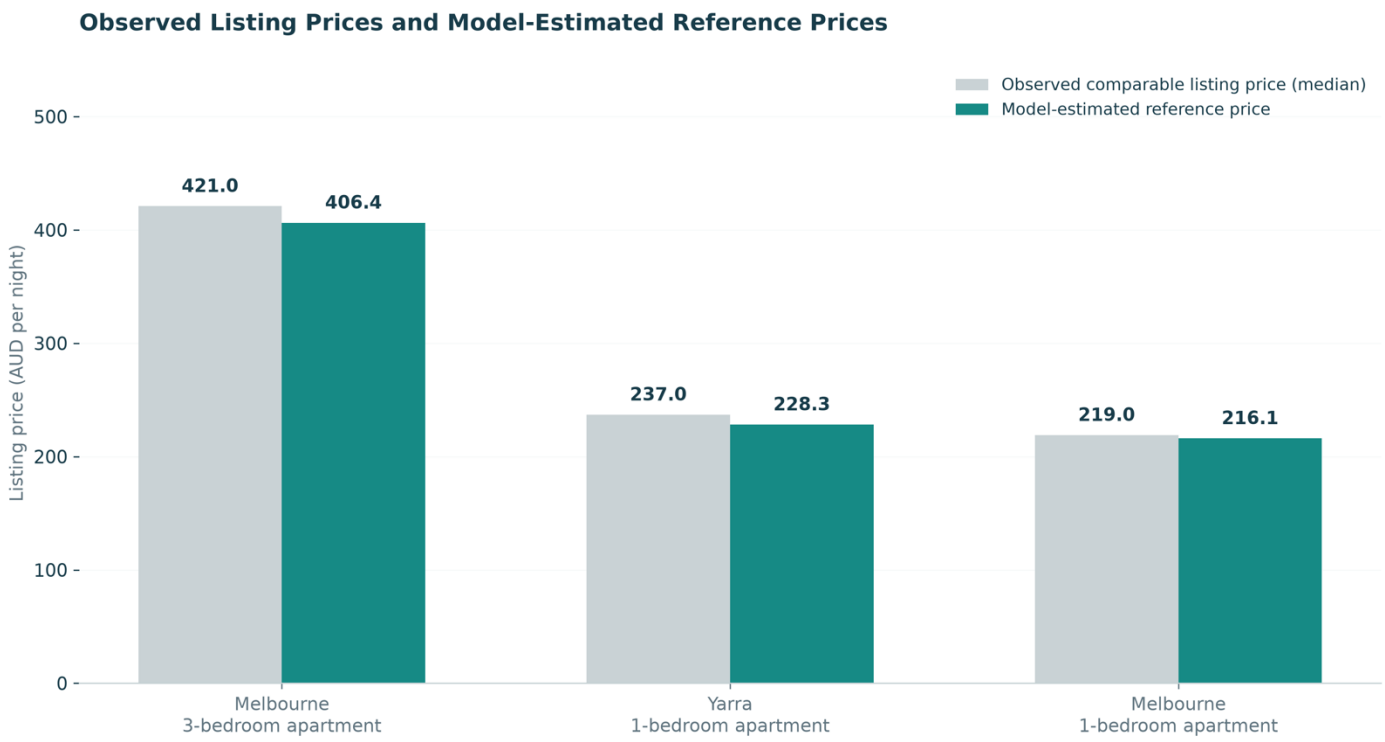


Figure 3

Observed comparable listing price is the outcome of simple pricing baseline, and model-estimated reference price is the output of the prediction model. Therefore, we get the best three products, and the output of the prediction model will be theirs best listing prices.

Prescriptive Analysis

This process will integrate the conclusions of the two analysis processes above and provide the final recommendation. This is done by applying explicit rules so each choice can be traced and repeated.

In order to convert the model predictions into initial quotations that are market-based and convenient to use, the observed listing price ranges of comparable properties must be examined for each of the three recommended property groups. Median model predictions serve as reference prices, while the 25th to 75th percentile range of observed property listing prices was used to provide a relatively focused range for the market quotations. According to the output (Table 1), all three model-estimated prices fall within comparable

listings' interquartile ranges and slightly below their medians. This indicates the model provides market-aligned starting prices. However, final pricing requires tracking views, inquiries, bookings, cancellations, and actual costs. Model estimates are therefore rounded to the nearest AUD 10 as initial suggested prices and scope for further adjustment.

Product (Range x bedrooms x type)	**25th – 75th percentile price (AU$)**	**Median (AU$)**	**Model reference (AU$)**
Melbourne 3b Apartment	347- 499	421	406.4
Yarra 1b Apartment	216- 262	237	228.3
Melbourne 1b Apartment	191- 249	219	216.1

Table 1

Based on the output, we get figure 4 as our specific priority product portfolio. The map shows the region that need to be taken into consideration and the specific top three products our client can operate first if they have one. Due to the selection condition includes the amount of the similar properties, The probability that our clients hold this type of property is relatively high. For the case that our client does not have such product or he/she has the product in the other areas, high-score-reviewed properties with multiple bedrooms that accommodate longer stays may also be suitable options for our clients, as the preceding analysis indicates that these characteristics are significantly associated with higher estimated revenue. Specific suggestions will be presented in the next section.

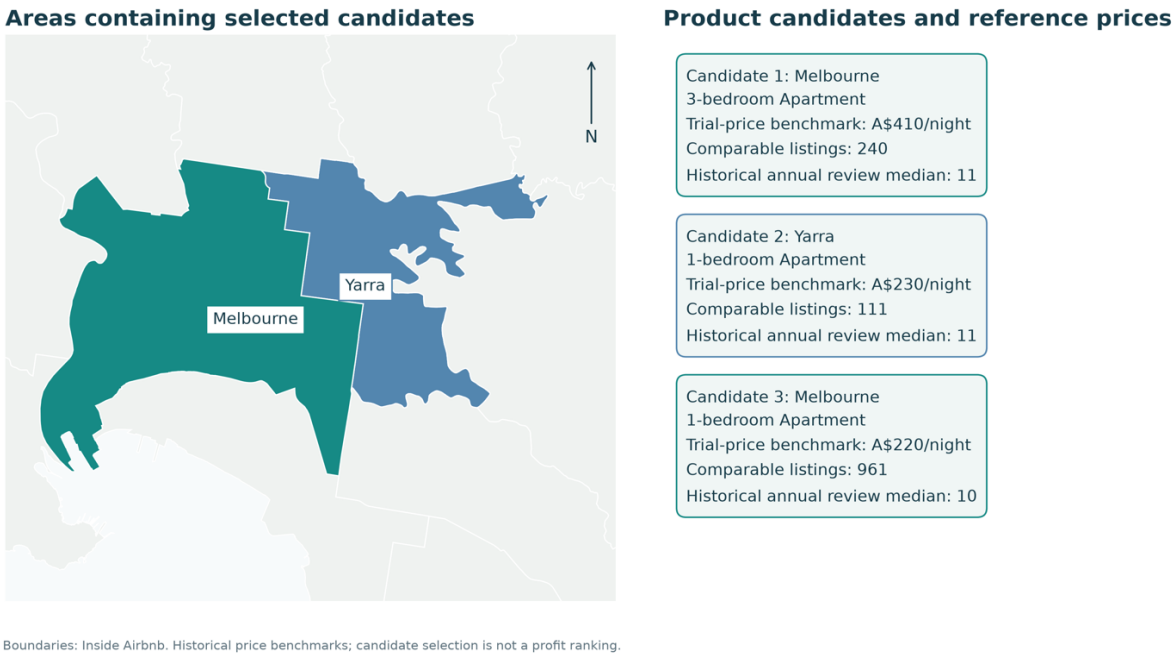


Figure 4

Recommendation

In conclusion, the findings from the analyses process provide answers to our four research questions. “Inner Metro” (Melbourne and Yarra) can be seen as premium area and three products in this area offer the strongest evidence base. The combination of market ranges and pricing model, it gives three launch options, including Melbourne three-bedroom, Yarra one-bedroom, and Melbourne one-bedroom at the starting listing price $410, $230, and $220 respectively. The larger units sell value per guest and the ratings are important. This means generally larger units with higher rate listings can easily earn more value.

Therefore, we recommend that clients prioritise the three property types identified above. Additionally, the client should focus more on service not just the price. This is because ratings help to bring higher revenue, therefore service quality is levering the client controls. It is suggesting using accurate photos, disclose all fees, give simple check-in and instructions, respond quickly, recurring complaints and make sure all amenities are

in good condition. Practical measures include thoroughly cleaning the property, accurately presenting the space through representative photographs and measurements, clearly disclosing all fees and facility restrictions, and reviewing prices against comparable listings for the same dates. Client could also ask Airbnb to test the entry key or procedure using the provided instructions and ensure that check-in handover, equipment instructions, and contact details are complete and clear. Purchasing appliances such as air conditioners and air purifiers requires additional investment, which falls outside the scope of our analysis. Although these upgrades may increase rental revenue, clients should assess whether the potential gains justify the purchase and ongoing operating costs.

Limitation and Future Plan

Although we have posted relatively feasible suggestions, our analysis still has limitations. Some are caused by the database itself, and some are due to the methods. The clients we are dealing with are non-superhosts, however, non-superhosts do not necessarily represent new operators. Our assumptions may not be rigorous. The data shows asking prices in June 2026 there is no transactions, occupancy or costs, therefore the recommend prices may not be able to maximize the profit.

Moreover, the review counts do not reflect as bookings, and higher review activity does not necessarily indicate greater profitability. There are some listings are lack of price or bedroom data, and the analysis focuses on mainstream entire-property short stays, so the findings cannot be directly extended to long-term, luxury, or other specialised markets. Furthermore, annual income is estimated rather than derived from actual statistics, so our database may not be able to conduct analysis based on the actual leasing situation. Therefore, the research provides room type selection and initial pricing references based on historical data. The actual effect

still needs to be verified and adjusted through the booking, transaction prices and cost data after going online.

For the method, the typical error in the model is about AUD 51 per night, so comparable-market checks and manual review are still needed. Feature importance does not establish causality. The analysis initially aimed to identify the most suitable properties in both central and suburban areas. However, the uneven distribution of listings generates that highly rated properties were concentrated in the city center. Consequently, our approach did not fully address the risk of generalizing city-dominated findings to suburban markets.

For the future, a one-month plan is formulated for our clients to collect real data and reduce the errors that may be caused by our database. In the first week, the client should confirm the property's location, capacity amenities costs and update the comparable prices for the same dates. The next two week, Airbnb should collect the transaction and cost over time, makes sure the reviews can be validated as the demand proxy and investigate missing prices and bedrooms. In the final week, when the client has detailed data of booking and cost evidence, the project can further study how demand that can affect its price. Then it can compare the minimum stay and improve on dynamic pricing and profit optimisation.

For the later research, the model can be refined based on the new data to better reflect actual operating conditions and incorporate appliance purchase costs This might enable more practical and better-informed recommendations.

Reference

Department of Transport and Planning. (n.d.). *Melbourne's future planning framework*. Victorian Government. https://www.planning.vic.gov.au/guides-and-resources/strategies-and-initiatives/melbournes-strategic-planning-history/melbournes-future-planning-framework

Inside Airbnb. (2026, June 16). *Melbourne, Victoria, Australia* [Data set]. https://insideairbnb.com/get-the-data/

Sun, Q., & Le, L. (2026). *CMCE30005-Renascence* [Source code]. GitHub. https://github.com/Qihang-cpu/MCE30005-Renascence

Appendix

Appendix A: Data sources

Dataset	Size
Listings	25,728 rows, 90 variables
Reviews	1,026,690 rows
Calendar	9,390,720 rows

Appendix B – Predictive model details

B1: Model selection (validation set)

Model	MAE	MedAE	Within 25%
Ridge regression	79.36	52.16	60.5%
Random forest	76.10	46.51	61.8%
Gradient boosting	**74.89**	**46.91**	**63.4%**

Note: Gradient boosting was selected because it achieved the lowest validation MAE.

Appendix C – Recommendation

C1: Illustrative monthly gross income (AU$)

Product (Starting price in $)	10 nights	15 nights	20 nights
Melbourne 3-bed (410)	4100	6150	8200
Yarra 1-bed (230)	2300	3450	4600
Melbourne 1-bed (220)	2200	3300	4400

Note: Starting price x booked nights. Not a revenue forecast or profit estimate; excludes fees, cleaning, rent and other costs.

C2: Price test levels (about 10% for each side, rounds to AU$10)

Product	Lower	Starting	Higher
Melbourne 3-bed room	370	410	450
Yarra 1-bedroom	210	230	250
Melbourne 1-bedroom	200	220	240